

how does AI harm the environment in a way that just using a lot of electricity doesn't?

Posted by u/keegballz • 0 points • 5 comments

just now i saw a post about how ai chatbots are drying places up by using water to run data centers. i've also generally seen the sentiment that it is bad for the environment. i get that the water makes the electricity, but i don't understand how ai uses up more water than anything else you do on your computer. wouldn't anything that uses a lot of electricity made with water be harmful?

Comments

u/TanukiThing • 1 points • 2025-05-24 03:59 UTC

This really breaks down into 2 issues: energy usage and refrigerant usage.

Generative AI uses a lot of power in two different ways: training and generating.

Training is a massively computationally intensive task, which took Meta 1.7 million GPU hours to train a single model. This requires an enormous amount of energy and millions of dollars of investment.

Actually using something like an LLM uses a lot of energy too. Figures I found online put it at about 10x the energy consumption of a Google search, though I've heard different figures thrown around. The real issue here is the *scale* at which we're using LLMs. They're built into everything nowadays, you can't make a search without also querying an AI agent. Websites also love to integrate them for no reason into every page. We're making an unfathomably large amount of LLM requests. Using generative AI for stuff like image generation is largely the same, but with a lower volume but much higher energy cost.

The other issue is refrigerants. To cool large data centers like those used to power AI models we have to use something called a refrigerant, which are notorious for their environmental impact. One common type of refrigerant are hydrofluorocarbons (HFCs). These are very similar to CFCs, which caused the Ozone hole, but contain hydrogen instead of chlorine. This means that HFCs do not damage the ozone, but have a greater impact on climate change. There are newer refrigerants that have less impact on climate change, but from my research it seems like the majority of data centers use HFCs.

u/Own_Active_1310 • 1 points • 2025-05-24 03:57 UTC

It doesn't, but the energy used by AI is substantially higher than the alternative methods we were using to answer questions.

That said, search engines and the Internet have declined in quality to the point where you basically need an AI to sift thru the digital landfills now.

u/ILikeJasmineRice • 1 points • 2025-05-24 03:47 UTC

Cooling.

Water is used a lot of cooling incredibly high temperature computer things, and a lot hotter than what your cpu is running on the daily.

Hence, rather than having a shitty aluminum heating, they use large and very intricate water coolers to cool as much as possible.

u/XRay2212xray • 1 points • 2025-05-24 03:45 UTC

AI uses massive amounts of power and unlike other things that have grown slowly and already exist, its new and is creating huge demand for power.

u/hellshot8 • 2 points • 2025-05-24 03:43 UTC

it doesnt, the issue is that it uses a lot of electricity for basically nothing